

Answer Key for PRACTICE TEST II, 2025 – 2026

Class X

MATHEMATICS STANDARD (Code No. 041)

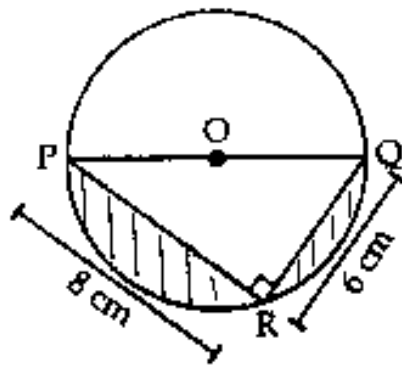
Time: 3 Hours

Maximum Marks: 80

SECTION A

Section A consists of 20 questions of 1 mark each.

- The LCM of the smallest prime and the smallest odd composite natural number is
(a) 6 (b) 12 (c) 18 (d) 24
- The perpendicular bisector of the line segment joining the points A(1, 5) and B(1, 21) cuts the y-axis at
(a) (0,13) (b) (0, -13) (c) (0,12) (d) (13,0)
- The value of k for which the pair of equations $x + y - 4 = 0$ and $2x + ky = 3$ will have no solution is
(a) - 2 (b) $\neq 2$ (c) 3 (d) 2
- If $d_1, d_2 (d_2 > d_1)$ be the diameters of two concentric circles and c be the length of the chord of a circle which is tangent to the other circle, then
(a) $d_2^2 = c^2 + d_1^2$ (b) $d_1^2 = c^2 + d_2^2$ (c) $c^2 = d_1^2 + d_2^2$ (d) $c = d_1 + d_2$
- If $\sin \theta - \cos \theta = 0$, then the value of $(\sin^4 \theta + \cos^4 \theta)$ is
(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
- Which one of the following is a quadratic equation?
(a) $(x + 1)^2 = (4 - x)^2 + 3$ (b) $-2x^2 = (5 - x) \left(2x - \frac{2}{5} \right)$
(c) $(k + 1)x^2 + \frac{3x}{2} = 7$ where $k = (-1)$ (d) $x^3 - x^2 = (x - 1)^3$
- In the figure, O is the center of the circle, PR and RQ are chords of the circle. The radius of the circle is 5 cm. PR = 8 cm, QR = 6 cm and $\angle PRQ = 90^\circ$. What is the approximate area of the shaded region?



- (a) $\left(\frac{25\pi}{4} - 24\right) \text{ cm}^2$ (b) $\left(\frac{25\pi}{2} - 24\right) \text{ cm}^2$ (c) $\left(\frac{25\pi}{4}\right) \text{ cm}^2$ (d) $\left(\frac{25\pi}{2}\right) \text{ cm}^2$

8. The probability that a non-leap year selected at random will contain 53 Sundays is

- (a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{5}{7}$

9. If $4 \tan \theta = 5$, then $\left(\frac{4 \sin \theta - \cos \theta}{4 \sin \theta + \cos \theta}\right)$ is equal to

- (a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

10. Prime divisors of the number $3^{13} - 3^{10}$ are

- (a) 2 and 3 only (b) 3 and 13 only
(c) 2 and 13 (d) 2, 3 and 13

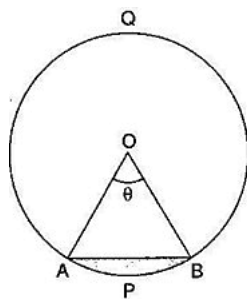
11. A solid is hemispherical at the bottom and conical (of same radius) above it. If the surface areas of the two parts are equal, then the ratio of its radius and the slant height of the conical part is

- (a) 1 : 2 (b) 2 : 3 (c) 2 : 5 (d) 3 : 2

12. For every nonzero real number p , the zeroes of the quadratic polynomial $x^2 + px + p$

- (a) cannot both be positive (b) cannot both be negative
(c) are always unequal (d) are always equal

13. In the following figure, O is the centre of the circle. The area of the sector OAPB is $\frac{5}{18}$ of the area of the circle. Then the value of θ is



- (a) 50° (b) 90° (c) 100° (d) 110°

14. If $\triangle ABC \sim \triangle EDF$, then which of the following is not true?

- (a) $BC \cdot EF = AC \cdot FD$ (b) $AB \cdot EF = AC \cdot DE$
 (c) $BC \cdot DE = AB \cdot EF$ (d) $BC \cdot DE = AB \cdot FD$

15. A school has 5 houses A, B, C, D and E. A class has 23 students: 4 from house A, 8 from house B, 5 from house C, 2 from house D and rest from house E. A student is selected at random from the class to be the class monitor. The probability that the selected student is not from A, B and C is

- (a) $\frac{4}{23}$ (b) $\frac{6}{23}$ (c) $\frac{8}{23}$ (d) $\frac{17}{23}$

16. The midpoint of the line segment joining the points $P(-4,5)$ and $Q(4,6)$ lies on

- (a) x-axis (b) y-axis (c) origin (d) one of the quadrants

17. The mean of m terms is $\bar{x}$. If the first term is increased by 1, second term by 2 and so on, then the new mean is

- (a) $\bar{x} + \frac{m-1}{2}$ (b) $\bar{x} + \frac{m+1}{2}$ (c) $\bar{x} + \frac{m}{2}$ (d) $\bar{x} + m$

18. If two tangents inclined at an angle 60° are drawn to a circle of radius 3 cm, then the length of each tangent is

- (a) $\frac{3}{2}\sqrt{3}$ cm (b) 6 cm (c) 3 cm (d) $3\sqrt{3}$ cm

Direction : In the question number 19 and 20, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**.

Choose the correct option :

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

- (b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.
19. **Assertion (A)** : If $\text{HCF}(105, 225) = 15$ and $\text{LCM}(105, 225) = 225 \times k$, then the value of k is 7.

Reason (R) : For any two positive numbers a and b , $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$.

Ans : (a)

20. **Assertion (A)**: For $0 < \theta \leq 90^\circ$, $\text{cosec } \theta - \cot \theta$ and $\text{cosec } \theta + \cot \theta$ are reciprocal to each other.

Reason (R): $\text{cosec}^2 \theta - \cot^2 \theta = 1$

Answer: (a)

SECTION B

Section B consists of 5 questions of 2 marks each.

21. **(A) Which term of the progression 65, 61, 57, 53,..... is the first negative term ?**

Answer : The progression 65, 61, 57, 53, is an AP as the difference between two consecutive terms is same throughout.

$$\therefore d = 61 - 65 = -4$$

$$a = 65$$

$$a_n = a + (n - 1)d = 65 + (n - 1)(-4) = 69 - 4n$$

$$\text{Now, } a_n < 0$$

$$\Rightarrow 69 - 4n < 0$$

$$\Rightarrow 4n > 69$$

$$\Rightarrow n > 17\frac{1}{4}$$

$$\Rightarrow n = 18, 19, 20, \dots$$

Hence the first negative term is the 18th term.

OR

(B) If the sum of first n , $2n$ and $3n$ terms of an AP be S_1 , S_2 and S_3 respectively then prove that $S_3 = 3(S_2 - S_1)$.

Answer:

$$S_1 = \frac{n}{2} [2a + (n - 1)d],$$

$$S_2 = \frac{2n}{2} [2a + (2n - 1)d] \text{ and}$$

$$\text{L.H.S: } S_3 = \frac{3n}{2} [2a + (3n - 1)d]$$

$$\text{R.H.S: } 3(S_2 - S_1)$$

$$= 3 \left[\{2na + n(2n - 1)d\} - \left\{na + \frac{1}{2}n(n - 1)d\right\} \right]$$

$$= 3 \left[na + \frac{3}{2}n^2d - \frac{1}{2}nd \right]$$

$$= \frac{3n}{2} [2a + 3nd - d]$$

$$= \frac{3n}{2} [2a + (3n - 1)d] = S_3 .$$

Hence, L.H.S = R.H.S

22. If $x \tan 30^\circ + y \cot 60^\circ = 0$ and $2x - y \tan 45^\circ = 1$, calculate the values of x and y .

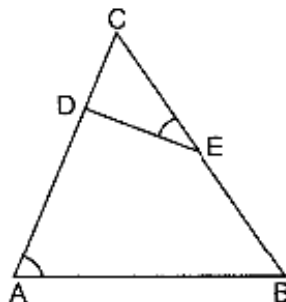
Answer:

$$x \tan 30^\circ + y \cot 60^\circ = 0 \Rightarrow x \times \frac{1}{\sqrt{3}} + y \times \frac{1}{\sqrt{3}} = 0 \Rightarrow x + y = 0 \text{ ----(I)}$$

$$2x - y \tan 45^\circ = 1 \Rightarrow 2x - y \times 1 = 1 \Rightarrow 2x - y = 1 \text{ -----(II)}$$

$$\text{Solving (I) \& (II), } x = \frac{1}{3} \text{ and } y = -\frac{1}{3}$$

23. In the following figure, if $\angle CAB = \angle CED$, then prove that $AB \times DC = ED \times BC$.



Answer:

In $\triangle CAB$ and $\triangle CED$,

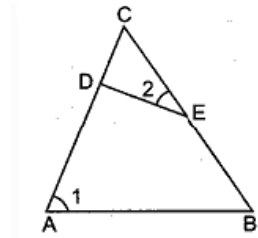
$$\angle CAB = \angle CED$$

$\angle C$ is common

$\therefore \triangle CAB \sim \triangle CED$ [By AA similarity]

$$\therefore \frac{BC}{DC} = \frac{AB}{ED}$$

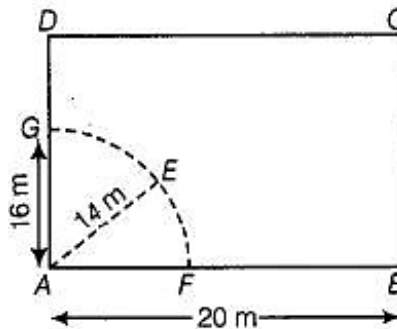
$$\Rightarrow AB \times DC = ED \times BC. \text{ (Proved)}$$



24. (A) A cow is tied with a rope of length 14 m at the corner of a rectangular field of dimensions 20 m $\times$ 16 m. Find the area of the field in which the cow can graze.

Answer :

Let ABCD be a rectangle of dimensions 20 m $\times$ 16 m. Suppose a cow is tied at a point A. Let the length of the rope be AE.



$$\therefore AE = r = 14 \text{ m}$$

$\therefore$ Area of the field in which the cow can graze

$$= \text{area of sector AFEG} = \frac{\theta}{360} \pi r^2$$

$$= \frac{90}{360} \times \frac{22}{7} \times (14)^2 \text{ m}^2 [\because \angle A = 90^\circ]$$

$$= 154 \text{ m}^2$$

OR

- (B) The area of a sector of central angle 200° of a circle is 770 cm^2 . Find corresponding arc of this sector.

Answer:

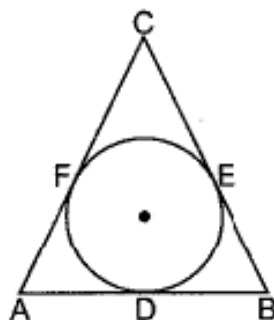
Let the central angle (in degrees) be θ .

$$\text{By the problem, } 770 = \frac{200}{360} \times \frac{22}{7} \times r^2$$

$$\Rightarrow r = 21$$

$$\text{Length of the arc} = \frac{200}{360} \times 2 \times \frac{22}{7} \times 21 = 73\frac{1}{3} \text{ cm}$$

25. In the following figure, a circle is inscribed in a $\triangle ABC$, such that it touches the sides AB, BC and CA at points D, E and F respectively. If the lengths of sides AB, BC and CA are 12 cm, 8 cm and 10 cm respectively, find the lengths of AD, BE and CF.



Answer:

AB = 12 cm, BC = 8 cm, CA = 10 cm

Let AD = AF = x [$\because$ tangents drawn from an external point are equal]

$\therefore$ DB = BE = 12 - x and CF = CE = 10 - x [$\because$ tangents drawn from an external point are equal]

Now, BC = 8 cm

$$\therefore \text{BE} + \text{CE} = 8 \text{ cm}$$

$$\Rightarrow 12 - x + 10 - x = 8$$

$$\Rightarrow 2x = 14$$

$$\Rightarrow x = 7$$

$$\therefore \text{AD} = 7 \text{ cm, BE} = (12 - 7) \text{ cm} = 5 \text{ cm and CF} = (10 - 7) \text{ cm} = 3 \text{ cm}$$

SECTION C

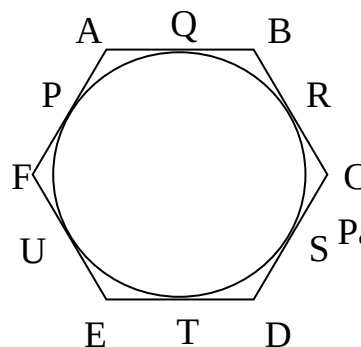
Section C consists of 6 questions of 3 marks each.

26. If a hexagon ABCDEF circumscribes a circle, prove that AB + CD + EF = BC + DE + FA.

Answer :

Given : A hexagon ABCDEF is circumscribed in a circle.

To prove : AB + CD + EF = BC + DE + FA



$$\begin{aligned}\text{Proof : } AB + CD + EF &= (AQ + QB) + (CS + DS) + (EU + FU) \\ &= (AP + BR) + (CR + DT) + (ET + FP)\end{aligned}$$

[$\because$ tangents from an external point are equal]

$$\begin{aligned}&= (AP + FP) + (BR + CR) + (DT + ET) \\ &= AF + BC + DE\end{aligned}$$

$$\therefore AB + CD + EF = BC + DE + FA$$

27. **On a morning walk, three persons step off together and their steps measure 40 cm, 42 cm and 45 cm respectively. What is the minimum distance each should walk, so that each can cover the same distance in complete steps ?**

Answer :

The minimum distance each should walk = LCM of 40 cm, 42 cm and 45 cm

$$40 = 2 \times 2 \times 2 \times 5$$

$$42 = 2 \times 3 \times 7$$

$$45 = 3 \times 3 \times 5$$

$$\therefore \text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 = 2520$$

Thus, the minimum distance each should walk is 2520 cm = 25.20 m.

28. **If α and β are the zeroes of quadratic polynomial $f(x) = kx^2 + 4x + 4$ such that $\alpha^2 + \beta^2 = 24$, find the values of k .**

Answer:

$$\alpha + \beta = -\frac{4}{k} \text{ and } \alpha\beta = \frac{4}{k}$$

$$\alpha^2 + \beta^2 = 24 \Rightarrow (\alpha + \beta)^2 - 2\alpha\beta = 24$$

$$\Rightarrow \left(-\frac{4}{k}\right)^2 - 2 \times \frac{4}{k} = 24$$

$$\Rightarrow 3k^2 + k - 2 = 0$$

$$\Rightarrow (k + 1)(3k - 2) = 0$$

$$\Rightarrow k = -1 \text{ or } k = \frac{2}{3}$$

29. **(A) Prove that $(\sin \alpha + \cos \alpha)(\tan \alpha + \cot \alpha) = \sec \alpha + \operatorname{cosec} \alpha$.**

Answer :

$$\text{LHS} = (\sin \alpha + \cos \alpha)(\tan \alpha + \cot \alpha)$$

$$\begin{aligned}
&= (\sin \alpha + \cos \alpha) \left(\frac{\sin \alpha}{\cos \alpha} + \frac{\cos \alpha}{\sin \alpha} \right) \\
&= (\sin \alpha + \cos \alpha) \left(\frac{\sin^2 \alpha + \cos^2 \alpha}{\sin \alpha \cos \alpha} \right) \\
&= (\sin \alpha + \cos \alpha) \left(\frac{1}{\sin \alpha \cos \alpha} \right) \\
&= \frac{1}{\cos \alpha} + \frac{1}{\sin \alpha} \\
&= \sec \alpha + \operatorname{cosec} \alpha = \text{RHS}
\end{aligned}$$

OR

29. (B) Let ABCD be a square, P and Q are any two points on BC and CD respectively such that AP = 4 cm, PQ = 3 cm, AQ = 5 cm. The length of the side of the square is $\frac{16}{\sqrt{x}}$ cm. Find x.

Answer:

Since $5^2 = 3^2 + 4^2$, $\angle APQ = 90^\circ$

(by converse of Pythagoras Theorem)

In $\triangle APB$, $AB = 4 \cos \theta$ and $PB = 4 \sin \theta$

In $\triangle PCQ$, $PC = 3 \cos \theta$

Now, $AB = BC$

$$\Rightarrow AB = BP + PC$$

$$\Rightarrow 4 \cos \theta = 4 \sin \theta + 3 \cos \theta$$

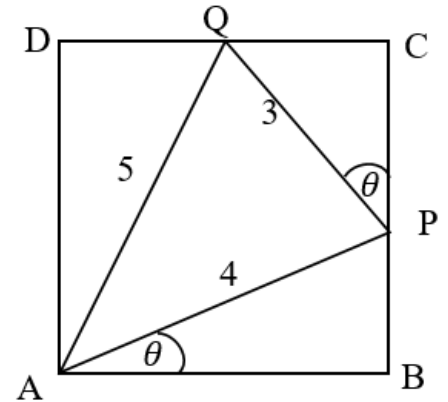
$$\Rightarrow \cos \theta = 4 \sin \theta$$

$$\Rightarrow \tan \theta = \frac{1}{4}$$

$$\Rightarrow \cos \theta = \frac{4}{\sqrt{17}}$$

$$\therefore AB = 4 \cos \theta = 4 \times \frac{4}{\sqrt{17}} = \frac{16}{\sqrt{17}} \text{ cm}$$

$$\therefore x = 17$$



30. From a pack of 52 cards, Jacks, Queens and Kings of red colour are removed. From the remaining, a card is drawn at random. Find the probability that the drawn card is:

- (i) A black king
- (ii) A card of red colour
- (iii) A card of black colour

Answer :

Number of cards removed = 6

$\therefore$ Remaining number of cards = $52 - 6 = 46$

(i) Number of black kings = 2

$$\therefore P(\text{a black king}) = \frac{2}{46} = \frac{1}{23}$$

(ii) Number of red colour cards remaining = $26 - 6 = 20$

$$\therefore P(\text{a card of red colour}) = \frac{20}{46} = \frac{10}{23}$$

(iii) Number of black cards = 26

$$\therefore P(\text{a card of black colour}) = \frac{26}{46} = \frac{13}{23}$$

31. (A) It can take 12 hours to fill a swimming pool using two pipes. If the pipe of larger diameter is used for 4 hours and the pipe of smaller diameter is used for 9 hours, only half the pool can be filled. How long would it take for each pipe to fill the pool separately.

Answer:

Let the time taken by the pipe of larger diameter to fill the pool be x hours and that taken by the pipe of smaller diameter by y hours.

$$\text{BTP } \frac{4}{x} + \frac{9}{y} = \frac{1}{2} \Rightarrow 4u + 9v = \frac{1}{2} \text{ -----(I)}$$

$$\frac{12}{x} + \frac{12}{y} = 1 \Rightarrow 12u + 12v = 1 \text{ -----(II)}$$

$$\text{where } u = \frac{1}{x} \text{ and } v = \frac{1}{y}$$

$$\text{So } u = \frac{1}{20}, v = \frac{1}{30}$$

$$\Rightarrow x = 20, y = 30$$

Therefore, the pipe of larger diameter alone can fill the pool in 20 hours and the pipe of smaller alone can fill the pool in 30 hours.

OR

(B) Solve the following system of equations graphically: $2x + y = 4$ and $2x - y = 4$. Find the area of the triangle so formed by the two lines and y-axis.

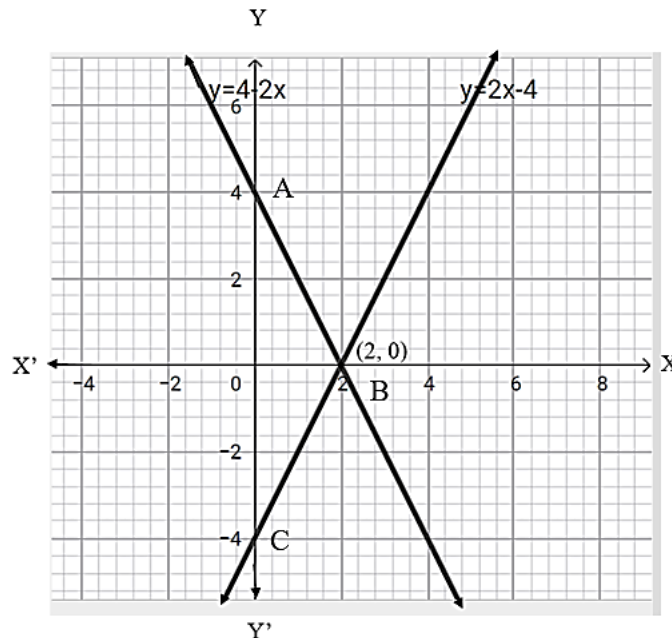
Answer : For the line $2x + y = 4$

x	0	2	4
y	4	0	-2

For the line $2x - y = 4$

x	0	2	4
y	-4	0	2

Graphical representation of the two lines:



From the graph, we see that $x = 2, y = 0$ as B (2 ,0) is their point of intersection.

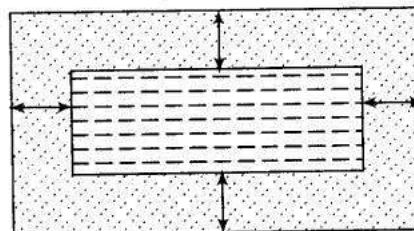
Area of the triangle formed by the lines and the y-axis

$$= \text{area of } \triangle ABC = \frac{1}{2} \times 8 \times 2 \text{ sq. units} = 8 \text{ sq. units}$$

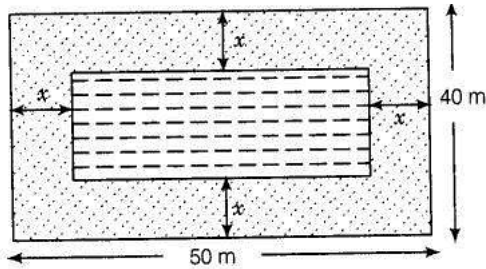
SECTION D

Section D consists of 4 questions of 5 marks each.

32. At the centre of a rectangular lawn of dimensions 50 m x 40 m, a rectangular pond has to be constructed as shown in the figure below, such that the area of the grass surrounding the pond would be 1184 m². Find the length and breadth of the pond.



Answer :



It is evident from the figure that the grass is of uniform thickness all around. Let the thickness be x m.

Length of the lawn = 50 m

Breadth of the lawn = 40 m

Length of the pond = $(50 - 2x)$ m

Breadth of the pond = $(40 - 2x)$ m

Area of the grass = 1184 m^2

$$\therefore 50 \times 40 - (50 - 2x)(40 - 2x) = 1184$$

$$\Rightarrow 2000 - 2000 + 180x - 4x^2 = 1184$$

$$\Rightarrow 4x^2 - 180x + 1184 = 0$$

$$\Rightarrow x^2 - 45x + 296 = 0$$

$$\Rightarrow x^2 - 37x - 8x + 296 = 0$$

$$\Rightarrow x(x - 37) - 8(x - 37) = 0$$

$$\Rightarrow (x - 37)(x - 8) = 0$$

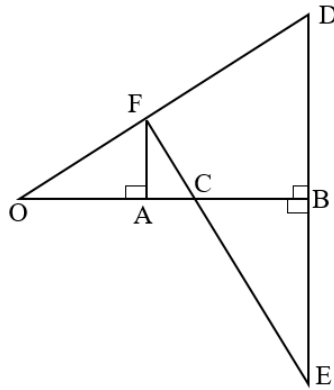
$$\Rightarrow x = 37 \text{ or } x = 8$$

If $x = 37$, then length of the pond = -24 m and its breadth = -34 m, but negative values are not possible.

$\therefore x = 8$ and hence length of the pond = 34 m and breadth of the pond = 24 m.

33. In the given figure, OB is perpendicular bisector of the line segment DE, FA is

perpendicular to OB and FE intersects OB at the point C. Prove that $\frac{1}{OA} + \frac{1}{OB} = \frac{2}{OC}$



Answer:

In $\triangle AOF$ and $\triangle BOD$, $\angle AOF = \angle BOD$ and $\angle OAF = \angle OBD = 90^\circ$

Therefore, $\triangle AOF \sim \triangle BOD$ (By AA similarity)

$$\Rightarrow \frac{OA}{OB} = \frac{FA}{DB} \text{ -----(I)}$$

In $\triangle FAC$ and $\triangle EBC$, $\angle FAC = \angle EBC = 90^\circ$ and

$\angle ACF = \angle BCE$ (vertically opposite angles)

Therefore, $\triangle FAC \sim \triangle EBC$ (By AA similarity)

$$\Rightarrow \frac{FA}{EB} = \frac{AC}{BC} \text{ or, } \frac{FA}{DB} = \frac{AC}{BC} \text{ -----(II)}$$

From (I) & (II),

$$\frac{OA}{OB} = \frac{AC}{BC} \Rightarrow \frac{OA}{OB} = \frac{OC - OA}{OB - OC}$$

$$\Rightarrow OA(OB - OC) = OB(OC - OA)$$

$$\Rightarrow 2OA \cdot OB = OB \cdot OC + OC \cdot OA$$

$$\Rightarrow (OB + OA)OC = 2OA \cdot OB$$

$$\Rightarrow \frac{1}{OA} + \frac{1}{OB} = \frac{2}{OC} \text{ (Dividing OC on both sides)}$$

34. (A) Water is flowing at the rate of 15 km/hr through a pipe of diameter 14 cm into a cuboidal pond which is 50 m long and 44m wide.

i. In what time will the level of water in the pond rise by 21 cm?

ii. What should be the speed of water if the rise in water level is to be attained in 1 hour ?

Answer: Water level is to rise by $h = 21 \text{ cm} = \frac{21}{100} \text{ m}$

$$\text{Volume of water in the pond} = 50 \times 44 \times \frac{21}{100} \text{ m}^3 = 462 \text{ m}^3$$

$$\text{Radius of the pipe} = 7 \text{ cm} = \frac{7}{100} \text{ m}$$

$$\text{Area of cross-section of pipe} = \frac{22}{7} \times \frac{7}{100} \times \frac{7}{100} = \frac{77}{5000} \text{ m}^2$$

Volume of water flowing in 1 hour

= Area of cross-section of pipe $\times$ height of water coming out of pipe

$$= \frac{77}{5000} \times 15000 \text{ m}^3$$

(i) Time required to fill the pond

$$= \frac{\text{volume of the pond}}{\text{volume of water flowing in 1 hour}}$$

$$= \frac{462}{\frac{77}{5000} \times 15000}$$

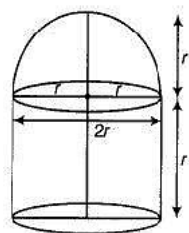
= 2 hours

(ii) Speed of water if the rise in water level is to be attained in 1 hour = (30 km/hr)

OR

(B) A building is in the form of a cylinder surmounted by a hemispherical vaulted dome and contains $41 \frac{19}{21} \text{ m}^3$ of air. If the internal diameter of the dome is equal to its total height above the floor, find the height of the building.

Answer : Let the total height of the building = internal diameter of the dome = $2r \text{ m}$



$\therefore$ Radius = $r \text{ m}$

Height of the cylinder = $(2r - r) \text{ m} = r \text{ m}$

$$\text{Volume of the cylinder} = \pi r^2 h = \pi r^2 (r) = \pi r^3 \text{ m}^3$$

$$\text{Volume of hemispherical dome} = \frac{2}{3}\pi r^3 \text{ m}^3$$

$$\therefore \text{Total volume of the building} = \pi r^3 \text{ m}^3 + \frac{2}{3}\pi r^3 \text{ m}^3 = \frac{5}{3}\pi r^3 \text{ m}^3$$

$$\text{So, } \frac{5}{3}\pi r^3 = 41\frac{19}{21}$$

$$\Rightarrow \frac{5}{3}\pi r^3 = \frac{880}{21}$$

$$\Rightarrow \frac{5}{3} \times \frac{22}{7} \times r^3 = \frac{880}{21}$$

$$\Rightarrow r^3 = \frac{880}{21} \times \frac{7}{22} \times \frac{3}{5}$$

$$\Rightarrow r^3 = 8$$

$$\Rightarrow r = 2$$

Thus the height of the building is $2r = 2 \times 2 = 4 \text{ m}$.

35. (A) If the mean of the following frequency distribution is 65.6, find the missing frequencies f_1 and f_2 .

Class	10 – 30	30 – 50	50 – 70	70 – 90	90 – 110	110 – 130	Total
Frequency	5	8	f_1	20	f_2	2	50

Answer:

Class Interval	Frequency	Class Mark	$f_i x_i$
10 – 30	5	20	100
30 – 50	8	40	320
50 – 70	f_1	60	$60f_1$
70 – 90	20	80	1600
90 – 110	$15 - f_1$	100	$1500 - 100f_1$
110 – 130	2	120	240
Total	50		$3760 - 40f_1$

From table we get, $f_2 = 15 - f_1$

$$\text{Also, } \frac{3760 - 40f_1}{50} = 65.6 \Rightarrow 40f_1 = 3760 - 3280 = 480 \Rightarrow f_1 = \frac{480}{40}$$

$$\Rightarrow f_1 = 12 \text{ and } f_2 = 3$$

OR

(B) If the median of the following frequency distribution is 32.5, find the values of p and q.

Classes	Frequency
0 – 10	p
10 – 20	5
20 – 30	9
30 – 40	12
40 – 50	q
50 – 60	3
60 – 70	2
Total	40

Answer :

Classes	Frequency	Cumulative Frequency
0 – 10	p	p
10 – 20	5	p + 5
20 – 30	9	p + 14
30 – 40	12	p + 26
40 – 50	q	p + q + 26
50 – 60	3	p + q + 29
60 – 70	2	p + q + 31
Total	40	

$$N = 40$$

$$\text{So } \frac{N}{2} = 20$$

Median is 32.5

$$\therefore \text{Median class} = 30 - 40$$

$$\text{Here } l = 30, f = 12, cf = p + 14, h = 10$$

$$\text{Now, Median} = l + h \left(\frac{\frac{N}{2} - cf}{f} \right)$$

$$\Rightarrow 32.5 = 30 + 10 \left(\frac{20-p-14}{12} \right)$$

$$\Rightarrow 3 = 20 - p - 14$$

$$\Rightarrow p = 3$$

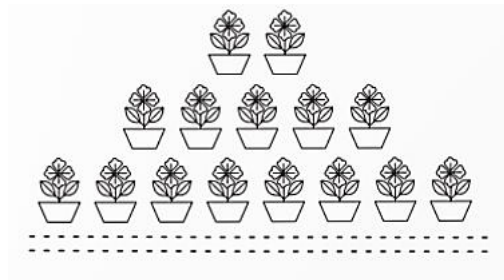
$$\text{Again, } p + q + 31 = 40$$

$$\Rightarrow q = 40 - 31 - 3 = 6$$

SECTION E

There are three Case study-based questions of 4 marks each which are compulsory.

36. Mihi being a plant lover decides to convert her balcony into a beautiful garden full of plants. She bought few plants with pots for her balcony. She placed the pots in such a way that the number of pots in the first row is 2, second row is 5, third row is 8 and so on.



Based on the above information, answer the following questions :

- (i) Find the number of pots placed in the 15th row. (1)
- (ii) Find the difference in the number of pots placed in 10th row and 5th row. (1)
- (iii) (a) If Mihi wants to place 100 pots in all, then find the total number of rows formed in the arrangement. (2)

OR

- (b) If Mihi has sufficient space for 12 rows, then how many pots in all can she place with this arrangement and what is the total cost if each pot costs Rs. 25? (2)

Answer :

- (i) Here the number of pots placed in successive rows are 2, 5, 8,

It forms an AP as the consecutive numbers differ by the same number 3.

So, $a = 2$ and $d = 3$

$$\therefore a_{15} = a + (15 - 1)d = 2 + 14 \times 3 = 44$$

(ii) We need to find $a_{10} - a_5 = a + (10 - 1)d - a - (5 - 1)d$

$$= 9d - 4d = 5d = 5 \times 3 = 15$$

(iii) (A) Let the total number of rows formed be n .

Here, $S_n = 100, a = 2, d = 3$

$$\therefore \frac{n}{2} \{2 \times 2 + (n - 1)(3)\} = 100$$

$$\Rightarrow \frac{n}{2} (3n + 1) = 100$$

$$\Rightarrow 3n^2 + n - 200 = 0$$

$$\Rightarrow 3n^2 + 25n - 24n - 200 = 0$$

$$\Rightarrow n(3n + 25) - 8(3n + 25) = 0$$

$$\Rightarrow (3n + 25)(n - 8) = 0$$

$$\Rightarrow n = 8 \text{ or } n = \frac{-25}{3}$$

$\therefore n$ cannot be negative or a fraction, so $n = 8$

Thus 8 rows will be formed.

OR

(B) Here $n = 12, a = 2, d = 3$

$$\therefore S_{12} = \frac{12}{2} \{2 \times 2 + (12 - 1)(3)\} = 6(4 + 33) = 222$$

222 pots can be placed.

$$\therefore \text{Total cost of pots} = \text{Rs. } 25 \times 222 = \text{Rs. } 5550$$

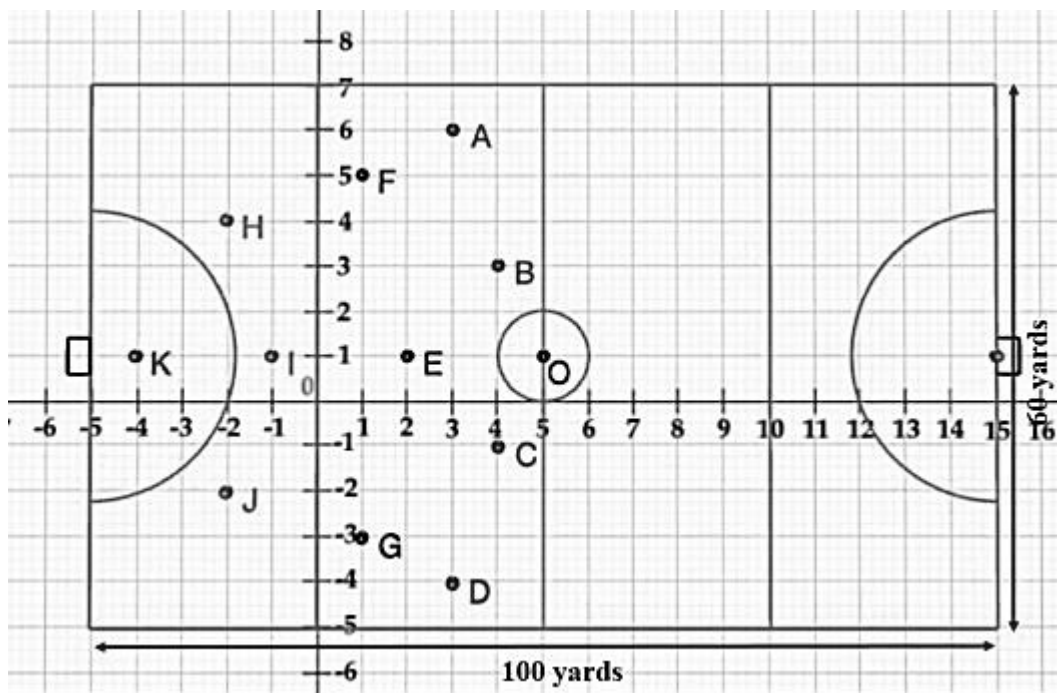
37. Vishal is the captain of the school hockey team. Position of every player includes

Forward: As shown by players A, B, C and D

Mid Fielders: As shown by players E, F and G

Full Backs: As shown by players H, I and J

Goal Keeper: As shown by player K



Based on the above information, answer the following questions.

- i. Find the coordinates of the position of the goalkeeper K. (1)
- ii. Find the distance between the center point of stadium i.e. O and the goalkeeper (1)
- iii. A) In which ratio does the y-axis divides the line segment joining the players represented by the points E and K. (2)

OR

- B) In which ratio does the x-axis divides the line segment joining by the players represented by F and G. (2)

Answer:

- i. The position of the goal keeper is K(-4,1).
- ii. The position of centre point of stadium is O(5,1).

Therefore, the required distance = $\sqrt{[(5 - (-4))]^2 + (1 - 1)^2} = 9$ units

- iii. (A) E(2,1) and K(-4,1)

Let the y-axis divide E and K in the ratio k:1

$$\therefore 0 = \frac{2 - 4k}{k + 1} \Rightarrow k = \frac{1}{2}$$

Therefore, the required ratio is 1 : 2.

OR

(B) F(1,5) and G(1, -3)

Let the x-axis divide F and G in the ratio $\lambda:1$

$$\therefore 0 = \frac{5-3\lambda}{\lambda+1} \Rightarrow \lambda = \frac{5}{3}$$

Therefore, the required ratio is 5 : 3.

38. There is a fire incident happens in a godown. The main door is locked so, the fireman is trying to enter the godown from the window. He places the ladder (at a point A) against the wall such that its top reaches the window as shown in the figure.

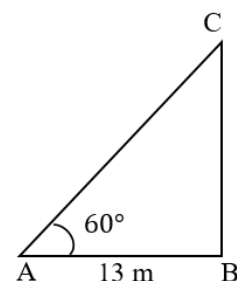


Based on the given information, answer the following questions.

- i. If fireman place the ladder 13m away from the wall and angle of elevation observed is 60° , then find the length of the ladder. (1)
- ii. If fireman place the ladder 13m away from the wall and angle of elevation observed is 60° , then find the height of the window. (1)
- iii. A) If the ratio of the height of the window and the godown is 2:3 and the angles of elevation from the point A to the window and the top of the godown are 30° and 45° respectively, then find the ratio of the corresponding length of ladders. (2)

OR

- iii. B) The angle of elevation from the point A to the window and top of the godown are 45° and 60° . Find the ratio of the heights of window and godown when the length of ladders are in the ratio 4:7. (2)



Answer:

$$(i) \cos 60^\circ = \frac{1}{2} = \frac{13}{AC} \Rightarrow AC = 26 \text{ m}$$

$$(ii) \tan 60^\circ = \sqrt{3} = \frac{BC}{13} \Rightarrow BC = 22.49 \text{ m}$$

(iii)(A) Let the height of window and the godown are $2x$ and $3x$.

From the figure,

$$\sin 30^\circ = \frac{BC}{AC} \Rightarrow AC = 4x \text{ ---(I)}$$

$$\text{and } \sin 45^\circ = \frac{BD}{AD} \Rightarrow AD = 3\sqrt{2}x \text{ ----(II)}$$

$$\text{from (I) and (II), } \frac{AC}{AD} = \frac{2\sqrt{2}}{3}$$

Therefore, the ratio of the corresponding length of ladders is $2\sqrt{2}:3$

OR

(B) Let the length of ladders are $4x$ and $7x$.

From the figure,

$$\sin 45^\circ = \frac{BC}{AC} \Rightarrow BC = 2\sqrt{2}x \text{ ---(I)}$$

$$\text{and } \sin 60^\circ = \frac{BD}{AD} \Rightarrow BD = \frac{7\sqrt{3}x}{2} \text{ ----(II)}$$

$$\text{from (I) and (II), } \frac{BC}{BD} = \frac{4\sqrt{2}}{7\sqrt{3}}$$

Therefore, the ratio of the heights of window and godown is $4\sqrt{2}:7\sqrt{3}$

